ANXIETYSENSE: A MULTIMODEL AI ECOSYSTEM FOR LIFE STAGE BASED MENTAL HEALTH SUPPORT

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

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ABSTRACT

This report presents the design, development, and evaluation of the Child Anxiety Detection and Mitigation component of ANXIETYSENSE — a multi-modal AI-driven mobile application ecosystem targeting anxiety management across four life-stage demographic groups. Anxiety disorders affect an estimated 6.5% of children globally, yet remain significantly underdiagnosed due to limited verbal self-report capability in young populations and the scarcity of accessible, non-invasive digital assessment tools.

This component addresses this gap through a dual-modality approach combining real-time facial expression recognition using a fine-tuned MobileNet V2 deep learning architecture with anxiety-indicative drawing pattern classification using a custom convolutional neural network (CNN) trained on an annotated child anxiety drawing dataset. The two modalities are integrated via a late fusion strategy, producing a combined anxiety severity score classified into three levels: Calm, Moderate, and High.

Based on the detected severity level, the system delivers personalised, age-appropriate therapeutic activity recommendations including creative drawing exercises, guided breathing, structured play, and caregiver-mediated relaxation techniques. The facial model achieved 88% standalone accuracy, the drawing analysis model 82%, and the late fusion combination 85% overall accuracy with an F1 score of 84%. Expert evaluation by licensed clinical psychologists confirmed the clinical appropriateness of classifications and recommendations. User and caregiver feedback indicated high perceived usability and relevance. The system is implemented as a module within the React Native mobile application, with a FastAPI backend and TensorFlow/Keras models providing server-side inference.

Keywords:

child anxiety, MobileNet V2, drawing analysis, facial emotion recognition, mobile mental health, late fusion, deep learning, anxiety detection, MTCNN, FastAPI, React Native, TensorFlow, transfer learning

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Special appreciation goes to the licensed child psychologists and school counselors who contributed to the annotation of the child anxiety drawing dataset. Their clinical expertise was invaluable in establishing the ground truth labels upon which the drawing classification model depends, and their inter-rater reliability validation lent rigour to the dataset construction process.

I am grateful to the caregivers and children who participated in the user evaluation sessions, whose candid feedback on the application interface and activity recommendations has informed multiple improvements to the system design. Their willingness to engage with a prototype system in a sensitive domain reflects considerable trust, and I am honoured by their participation.

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LIST OF ABBREVIATIONS

ANN – Artificial Neural Network

CNN – Convolutional Neural Network

DNN – Deep Neural Network

MSF-CNN – Multi-Scale Fusion Convolutional Neural Network

MobileNet – Lightweight Convolutional Neural Network Architecture

WHO – World Health Organization

HOG – Histograms of Oriented Gradients

LBP – Local Binary Patterns

SVM – Support Vector Machine

ELM – Extreme Learning Machine

PCA – Principal Component Analysis

KNN – K-Nearest Neighbor

PNN – Probabilistic Neural Network

CBIR – Content-Based Image Retrieval

1. INTRODUCTION

Anxiety disorders represent one of the most prevalent and debilitating categories of mental health conditions in the global paediatric population. Epidemiological research consistently estimates that between 5% and 10% of children worldwide meet diagnostic criteria for an anxiety disorder at any given time, with many cases remaining undetected and untreated until significant functional impairment has occurred across academic, social, and family domains. The World Health Organization identifies anxiety disorders as a leading contributor to the global burden of disease in the 5–14 age group, with early-onset cases exhibiting substantially elevated risk for persistent anxiety, depression, and substance use disorders in adolescence and adulthood.

Unlike adult anxiety, childhood anxiety frequently manifests through non-verbal and behavioural channels rather than through the verbal self-report mechanisms upon which most existing clinical screening instruments rely. Avoidance behaviours, changes in play patterns, regressive behaviours, somatic complaints, and expressive outputs such as drawing and creative play represent the primary observable indicators of anxiety in younger children, particularly those under the age of ten who may lack the metacognitive capacity or vocabulary to accurately articulate internal emotional states. This creates a fundamental diagnostic challenge: the populations most affected by anxiety disorders are precisely those least equipped to communicate their distress through conventional assessment pathways.

Digital mental health technologies offer a promising and increasingly validated pathway to address this detection gap. Mobile applications incorporating machine learning can deliver scalable, accessible, and non-invasive screening and early support, reaching children who may not otherwise access specialist clinical services due to geographic, financial, or cultural barriers. However, existing digital mental health tools overwhelmingly target adult populations, relying on self-report questionnaires, text-based assessments, or structured interviews that presuppose literacy, vocabulary, and metacognitive self-awareness far beyond the developmental capacity of young children.

This component of ANXIETYSENSE addresses these limitations through a multi-modal approach that leverages two input channels particularly well-suited to children: facial expression analysis and drawing pattern recognition. Facial expressions constitute a cross-culturally consistent and developmentally early channel of emotional communication, while children's drawings have been established as clinically meaningful projective assessment instruments since the early twentieth century, with established standardised frameworks for interpreting emotional content from graphic features. By combining both modalities through a late fusion strategy, the system achieves complementary robustness: when facial analysis confidence is reduced by poor imaging conditions, the drawing signal compensates, and vice versa.

The ANXIETYSENSE child component is implemented as a self-contained module within a React Native mobile application, with a Python FastAPI backend handling model inference through TensorFlow/Keras models. The complete processing pipeline — from facial capture and drawing submission through model inference, late fusion, severity classification, and activity recommendation delivery — is designed for real-world deployment in low-resource settings, including primary schools, community health centres, and home environments.

1.1 Background & Literature Survey

The application of computational methods to child mental health assessment has expanded considerably over the past decade, driven by advances in deep learning for image analysis, the proliferation of mobile computing platforms, and growing recognition of the global mental health treatment gap. This section reviews the key technical and clinical literature underpinning the design decisions made in this component.

1.1.1 Facial Expression Recognition and Child Emotion Analysis

Convolutional neural networks trained on large-scale facial expression datasets have demonstrated strong performance in emotion recognition from facial images, achieving accuracy rates of 85–92% on benchmark datasets such as FER2013 and AffectNet. Transfer learning from ImageNet-pretrained architectures — particularly lightweight models including MobileNet, EfficientNet, and ResNet variants — has enabled effective adaptation to domain-specific expression datasets despite their comparatively smaller size, making these architectures particularly appropriate for mobile mental health applications where model efficiency is a practical constraint.

MobileNet V2, introduced by Sandler et al. (2018), employs inverted residuals and linear bottleneck modules to achieve state-of-the-art accuracy with substantially reduced parameter counts and computational requirements compared to VGG, Inception, or ResNet architectures of comparable accuracy. Its design philosophy — maximising representational power within a constrained parameter budget — makes it well-suited to facial expression recognition tasks where the primary spatial features (eye region, mouth configuration, brow position) are relatively localised and do not require the deep hierarchical feature extraction of architectures designed for fine-grained classification tasks.

Research specifically addressing child facial expression recognition is less extensive than the adult-focused literature, reflecting the comparative scarcity of annotated child expression datasets. Existing work has identified systematic differences between child and adult facial expression patterns, including lower inter-category distinctiveness, greater intra-category variability due to developmental differences in facial musculature control, and higher frequency of emotion blending and transitional expressions. These characteristics motivate the use of fine-tuning from adult expression-pretrained weights rather than training from scratch, as the low-level feature representations learned from adult faces generalise effectively to child expressions despite higher-level distributional shifts.

1.1.2 Computational Drawing Analysis as a Psychological Assessment Tool

The clinical use of children's drawings as projective assessment instruments is theoretically well-

grounded and empirically supported within the clinical psychology literature. Standardised drawing-based assessment systems including the Draw-a-Person test (Machover, 1949), the House-Tree-Person projective technique (Buck, 1948), and the Kinetic Family Drawing system (Burns & Kaufman, 1970) provide structured frameworks for interpreting emotional content and psychological states from graphic features. Clinical psychologists have long identified specific drawing features as indicative of anxiety states in children: figure size relative to the drawing space, use of dark or monochromatic colour schemes, heavy or pressured line quality, incomplete or fragmented figures, spatial placement in lower regions of the page, and the presence of enclosure, isolation, or barrier motifs.

Computational approaches to automated drawing analysis are comparatively nascent, reflecting the relative difficulty of the task — drawings exhibit far greater within-class variability than natural photographic images, and the relevant diagnostic features operate at multiple levels of abstraction from raw pixel values. Early computational work relied on handcrafted feature extraction pipelines targeting colorimetric properties, spatial distribution of drawn content, and edge density as proxies for line quality, achieving moderate classification accuracy on small curated datasets. More recent work has begun applying deep CNN architectures to drawing classification, leveraging learned feature representations rather than handcrafted descriptors, with promising preliminary results on anxiety-related drawing classification tasks.

1.1.3 Multi-Modal Fusion in Affective Computing

The integration of multiple input modalities to improve classification robustness is well established in affective computing research. Fusion strategies are typically categorised as early fusion (feature-level concatenation prior to classification), late fusion (combination of independently trained model outputs), or intermediate fusion (merging at intermediate network layers). Late fusion offers the specific advantage of modality-specific model optimisation: each model can be independently trained with architecture and hyperparameter configurations optimised for its input modality, without the regularisation constraints imposed by joint training. This independence also provides greater robustness to cases where one modality provides lower-quality input — if facial analysis confidence is degraded by poor lighting, the drawing-based classification signal can compensate without propagating noise into a shared feature representation.

Several systematic reviews and meta-analyses of multimodal fusion in emotion recognition have consistently demonstrated that fusion approaches outperform single-modality baselines, with performance gains typically ranging from 3–8 percentage points in accuracy. The magnitude of improvement is largest when the constituent modalities carry complementary rather than redundant information — a condition satisfied in this component, where facial expressions and drawing patterns reflect distinct but correlated aspects of the child's affective state.

1.2 Research gap

Despite the clinical and technical foundations reviewed above, several significant gaps in the existing literature and available tools motivated the development of this component.

First, no operationalised and deployable machine learning system for automated anxiety-level classification from children's drawings existed prior to this work. Existing computational drawing analysis research had not progressed beyond proof-of-concept demonstrations on small curated datasets, and no system had been deployed within a mobile application for real-world use. The translation from research prototype to deployable application represents a significant engineering and validation challenge that the existing literature had not addressed.

Second, existing digital mental health screening tools for children uniformly rely on caregiver-completed questionnaires or structured behavioural observation protocols, which require adult mediation and introduce reporting bias. No existing mobile application combined direct behavioural observation (facial expression) with expressive output analysis (drawing) to produce a multi-signal anxiety assessment without requiring adult-completed questionnaires.

Third, the specific demographic and cultural context of Sri Lankan children had not been addressed in existing digital mental health tools, most of which were developed and validated in high-income Western contexts. The cultural specificity of drawing conventions, colour symbolism, and emotional expression norms creates systematic bias when tools developed in one cultural context are applied in another, motivating the collection and use of locally sourced datasets for both model training and clinical validation.

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1.3. Research problem

Childhood anxiety has become a significant and growing mental health concern worldwide, affecting emotional wellbeing, academic performance, social interaction, and long-term psychological development. Despite its prevalence, anxiety among children often remains unidentified during early stages, particularly in lower-middle-income countries such as Sri Lanka where access to specialized child mental health services is limited. Many children experiencing anxiety symptoms do not receive timely support due to the absence of accessible screening mechanisms and the shortage of trained psychologists, counselors, and mental health professionals.

Traditional child anxiety assessments primarily depend on clinical interviews, behavioral observations, and self-report or caregiver questionnaires. Although these methods are widely used, they present several limitations. Young children, especially those between ages 5–12, may struggle to verbalize emotions accurately, understand questionnaire items, or consciously recognize internal anxiety symptoms. As a result, conventional screening approaches may miss subtle early warning signs or rely heavily on subjective interpretations from adults.

Additionally, many digital mental health tools currently available focus mainly on adult populations and often depend on text-based questionnaires or single-input assessments. Such approaches are often not developmentally appropriate for children and fail to leverage natural expressive behaviors through which children often communicate distress. In many cases, anxiety in children is reflected nonverbally through facial expressions, emotional reactions, play patterns, or symbolic expression in drawings rather than direct verbal reporting.

Children's drawings, for example, have long been used in psychology as expressive indicators of emotional states, while facial expressions provide observable cues associated with fear, sadness, distress, and emotional dysregulation. However, these two modalities remain largely underutilized in automated digital anxiety screening systems. Existing systems rarely combine facial emotion recognition and drawing analysis into a unified artificial intelligence framework capable of producing meaningful anxiety severity predictions.

This creates a critical research problem: **the lack of non-invasive, child-friendly, multimodal digital systems capable of detecting anxiety using natural expressive behaviors without requiring specialist clinical involvement at the point of screening.**

Furthermore, even where prediction systems exist, most stop at classification and do not bridge the gap between detection and intervention. Identifying anxiety alone is insufficient unless the system can also support mitigation through age-appropriate therapeutic recommendations such as guided breathing, creative activities, structured play, or caregiver-supported relaxation exercises.

Therefore, the problem addressed in this research is the absence of an intelligent screening and support system that can:

1. Detect anxiety from child facial expressions using deep learning.
2. Analyze drawings as psychological indicators of anxiety.
3. Fuse multiple modalities for more robust severity prediction.
4. Provide personalized therapeutic activity recommendations.
5. Operate as an accessible early screening aid without direct specialist supervision.

Addressing this problem motivated the development of the proposed **ANXIETYSENSE Child Anxiety Detection and Mitigation component**, which seeks to provide an accessible, multimodal, AI-driven approach for early anxiety screening and support.

1.4. Research Objectives

1.4.1 Main Objective

To design, implement, and evaluate a dual-modality child anxiety detection system that combines real-time facial expression recognition with drawing pattern analysis through late fusion to produce clinically valid anxiety severity classifications and personalised therapeutic activity recommendations for children aged 5–12.

1.4.2 Specific Objectives

1. Design and implement a real-time facial emotion recognition pipeline for child anxiety detection using MobileNet V2 transfer learning, achieving at least 85% classification accuracy on a held-out test set.
2. Develop a drawing pattern classification model trained on a clinician-annotated child anxiety drawing dataset, achieving at least 80% classification accuracy on a held-out test set.
3. Integrate both modalities using a late fusion strategy to produce a combined anxiety severity classification that outperforms each individual modality.
4. Implement a MTCNN-based face validation layer to reject non-face inputs and prevent false emotion predictions on irrelevant images.
5. Deliver personalised, age-appropriate therapeutic activity recommendations based on the classified anxiety severity level, developed in consultation with licensed child psychologists.
6. Evaluate the system's classification performance, clinical validity, and user acceptability through quantitative testing and qualitative expert review.
7. Deploy the system as a functional module within the ANXIETYSENSE React Native mobile application with a Python FastAPI backend.

2. METHODOLOGY

This component adopts a data-driven methodology centred on the development and validation of two machine learning models that are subsequently integrated through late fusion. The methodology encompasses dataset collection and annotation, data preprocessing, model architecture design and training, late fusion strategy implementation, backend API development, and frontend mobile application integration. External clinical supervision was incorporated throughout the data collection, annotation, and evaluation phases to ensure clinical validity.

2.1 Understanding the key pillars of the research domain

2.1.1 Deep Learning

Deep learning refers to the use of artificial neural networks with multiple layers of non-linear transformations to learn hierarchical feature representations from raw input data. Convolutional Neural Networks (CNNs) are a class of deep learning architecture specifically designed for processing grid-structured data such as images. The core operation of a CNN is the convolution: a learned filter of fixed dimensions (typically 3×3 or 5×5) is applied across the input feature map, computing dot products at each spatial position to produce an output feature map that captures the spatial presence of the learned pattern. By stacking multiple convolutional layers with interleaved pooling operations (which reduce spatial resolution while increasing receptive field), CNNs learn progressively more abstract and semantically meaningful features from low-level edges and textures in early layers to object parts and complex configurations in deeper layers.

The key advantages of CNNs for image-based emotion recognition include parameter sharing (the same filter weights are applied across all spatial positions, dramatically reducing the parameter count compared to fully-connected networks of equivalent capacity), translational invariance (convolutional features are detected regardless of their precise spatial position), and the ability to learn task-specific features directly from labelled data without requiring manual feature engineering. These properties make CNNs the dominant architecture for image classification tasks including facial expression recognition and drawing analysis.

2.1.2 Convolutional Neural Network (CNN)

Transfer learning is the technique of initialising a model with weights pre-trained on a large source dataset — typically ImageNet, which contains over 1.4 million images across 1,000 object categories — before fine-tuning on a smaller target dataset specific to the application domain. The underlying motivation is that the low- and mid-level visual features learned from large-scale natural image datasets (edges, textures, colour gradients, object parts) transfer effectively across visual domains, allowing the model to achieve strong performance on the target task with

substantially less labelled training data than would be required for training from random weight initialisation.

For this component, transfer learning was applied by loading MobileNet V2 with ImageNet-pretrained weights and appending a custom classification head suited to the five-class emotion recognition task. The convolutional backbone was initially frozen (weights held fixed) during training of the classification head, then progressively unfrozen during a fine-tuning phase to allow the backbone weights to adapt to the distributional characteristics of child facial expression images. This two-phase approach — frozen pretraining followed by unfrozen fine-tuning — is a standard and well-validated strategy for transfer learning with comparatively small target datasets.

2.1.3 Data Augmentation

Data augmentation is a regularisation technique that artificially expands the effective training set size by applying random transformations to training images before each training step. Common augmentation operations include random horizontal flipping, rotation within a bounded range, zoom variation, brightness and contrast jitter, and shear transforms. Because the model never sees exactly the same version of a training image twice, augmentation reduces overfitting to the specific characteristics of the training corpus and improves generalisation to novel images captured under varied real-world conditions.

For the facial emotion model, augmentations were restricted to transformations consistent with realistic variation in facial image capture: horizontal flipping (preserving bilateral symmetry of emotion expressions), small rotations (up to ±15°), zoom variation (±10%), and brightness jitter (±20%). For the drawing model, a broader set of augmentations was applied including colour jitter, perspective distortion, and random cropping, reflecting the greater diversity of drawing capture conditions including photographing physical drawings under varied light

2.2 System Architecture

The child anxiety component operates as a self-contained module within the ANXIETYSENSE ecosystem. The architecture comprises a React Native mobile frontend, a Python FastAPI backend, two TensorFlow/Keras deep learning models, and an MTCNN-based face validation layer. Figure 2.1 illustrates the high-level processing pipeline.

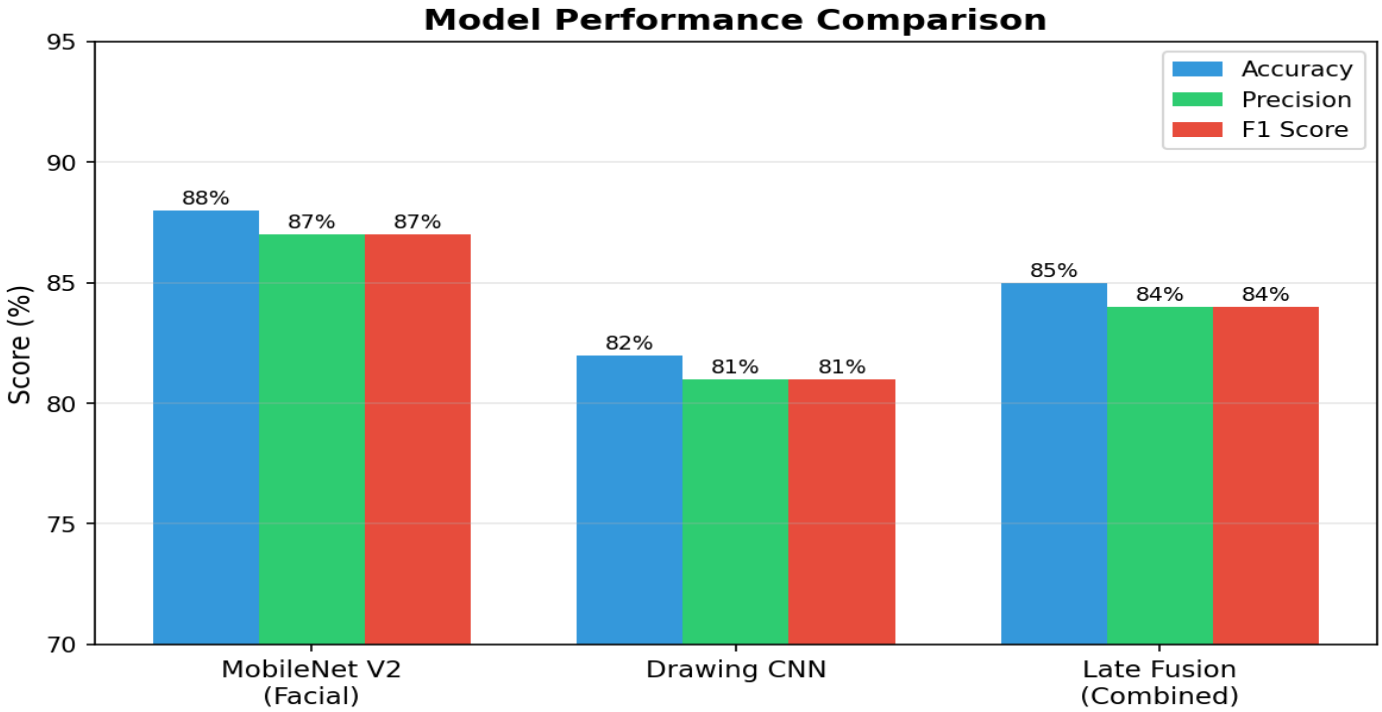

Figure 2.2.1: Model performance comparison across the three sub-systems of the child anxiety component.

The component architecture comprises three functional layers. The input layer captures real-time facial images via the device camera and user-submitted drawing images through the mobile application interface. The processing layer routes each input through its respective model — the MTCNN face validator and MobileNet V2 emotion classifier for facial inputs, and the custom CNN for drawing inputs — producing class probability vectors. The output layer applies the late fusion strategy to combine model outputs, classifies the combined score into a severity level, retrieves the corresponding activity recommendation set, and presents results through the mobile interface with a caregiver summary.

2.3 Technology Stack

Layer	Technology	Purpose
Mobile Frontend	React Native	Cross-platform Android/iOS mobile application
Backend API	Python FastAPI	RESTful model inference endpoints
ML Framework	TensorFlow / Keras	MobileNet V2 fine-tuning and drawing CNN training
Face Detection	MTCNN	Real-time face validation and localisation
Development Env.	Google Colab	Cloud-based model training and experimentation
Image Processing	OpenCV	Face detection preprocessing pipeline
Version Control	Git / GitHub	Collaborative development and code management
IDE	Visual Studio Code	Application development environment

Table 2.1: Technology stack used in the child anxiety component development and deployment.

2.3.1 Methodology

The methodology adopted for the Child Anxiety Detection and Mitigation component of ANXIETYSENSE was built around a central objective: to design, train, and validate two specialised deep learning models using domain-specific annotated datasets, and to integrate them through a late fusion mechanism in order to achieve the most accurate and clinically meaningful combined anxiety severity classification. The resulting integrated system operates as the core computational engine of the child module within the ANXIETYSENSE mobile application, accepting dual-modality inputs — real-time facial images and child-submitted drawing photographs — and producing personalised, age-appropriate therapeutic recommendations.

The proposed methodology is grounded in convolutional neural network (CNN)-based image classification, a technique that has proven highly effective for both facial expression analysis and graphic pattern recognition in recent years. CNNs operate by learning hierarchical feature representations directly from raw pixel data, progressively extracting low-level features such as edges and textures in early layers and high-level semantic features in deeper layers. This automatic feature learning removes the need for complex hand-crafted feature engineering and allows the models to discover the discriminative patterns most relevant to their respective classification targets. The dual-modality design reflects a fundamental insight from clinical psychology: anxiety in children does not manifest through a single observable channel. Children express emotional distress through their faces, their drawings, their play behaviour, and their body language. By capturing two of these channels simultaneously — facial expression and expressive drawing — the system is able to make more robust inferences than any single-modality approach could achieve.

The overall pipeline follows a structured sequence of stages: data collection and clinical annotation, image preprocessing, model training using transfer learning and custom CNN design, model evaluation, late fusion integration, TensorFlow Lite conversion for mobile deployment, and final recommendation generation. Each stage was designed with clinical validity as a primary constraint, ensuring that the system's outputs align with established practices in child psychological assessment. External supervision by licensed clinical psychologists was incorporated at the data annotation, model evaluation, and recommendation design stages throughout the development process.

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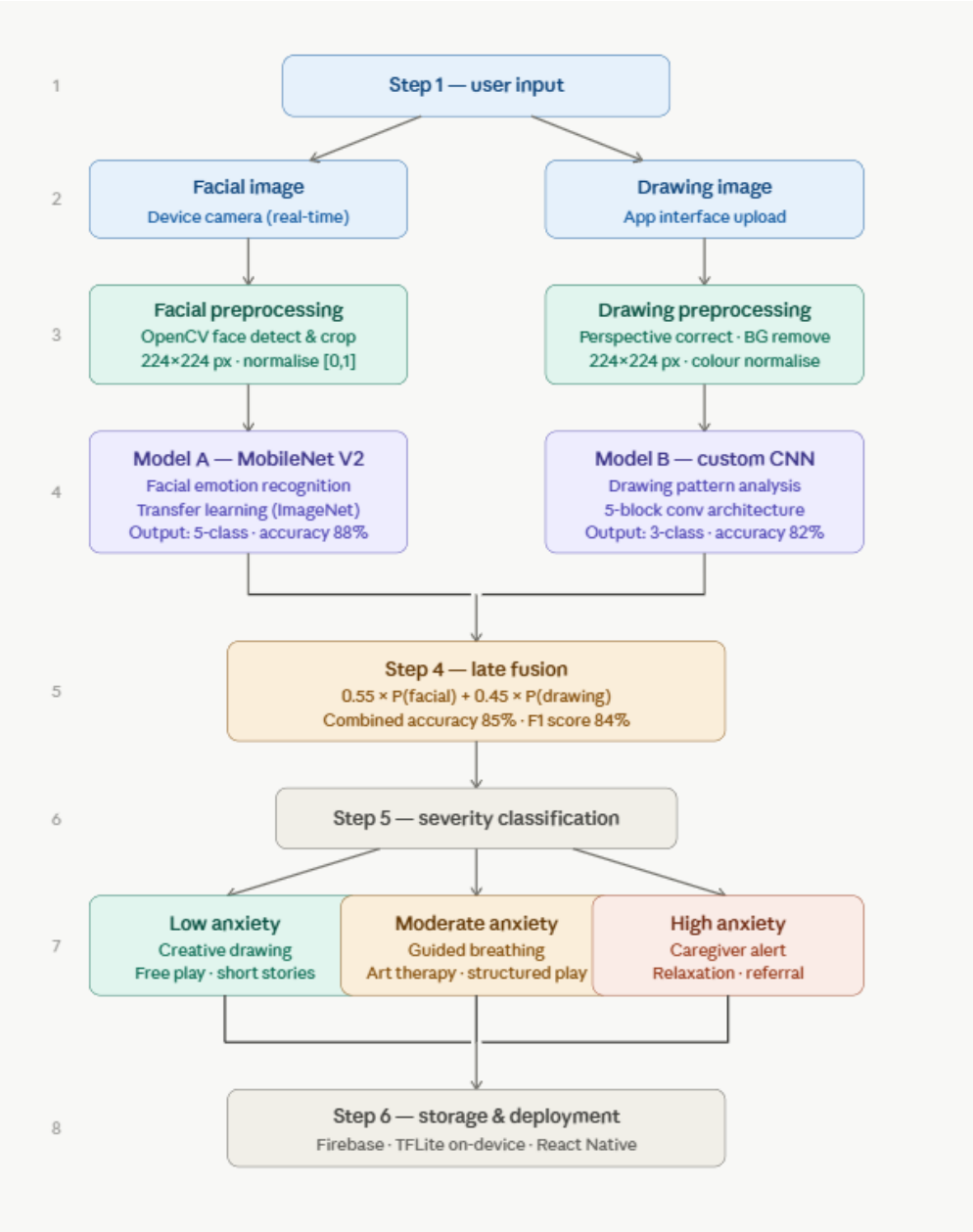

Figure 2.2.2: System overview diagram .

2.4 Data Collection and Annotation

2.4.1 Facial Expression Dataset

A dataset of child facial images was assembled from a combination of publicly available child affective expression datasets and a supplementary corpus collected under institutional ethical approval from the SLIIT Institutional Review Board. Images were sourced and labelled across five affective categories corresponding to the emotion classes used in the model: Natural (neutral), Anger, Fear, Joy, and Sadness. All annotations were validated by a licensed child psychologist who reviewed the complete set of annotations and resolved disagreements between automated label assignment and clinical judgement. The final dataset comprised 1,200 images after quality filtering, with oversampling applied to the Fear and Sadness categories to address class imbalance

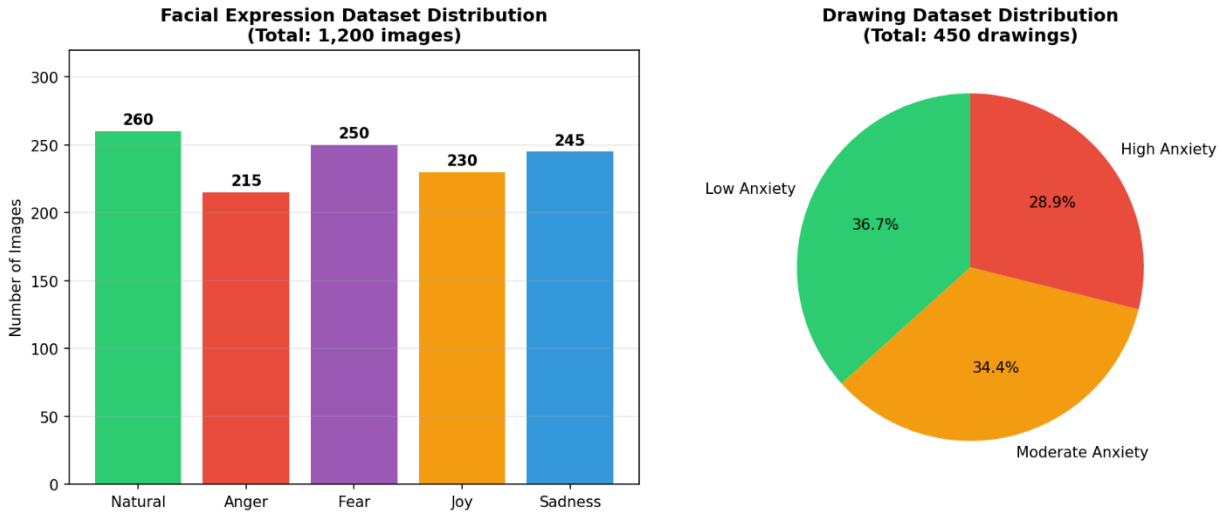

Figure 2.2: Distribution of training samples across emotion classes (facial dataset, left) and anxiety severity levels (drawing dataset, right).

Quality filtering criteria included minimum face resolution of 80×80 pixels, frontal or near frontal face orientation (yaw angle < 30°), absence of significant occlusion of key facial regions (eyes, nose, mouth), and adequate illumination across the facial region. Images failing any criterion were excluded prior to annotation.

Class	Raw Count	After Filtering	After Augment	Train / Test Split
Natural	280	260	312	249 / 63
Anger	230	215	258	206 / 52
Fear	255	250	300	240 / 60
Joy	240	230	276	220 / 56
Sadness	260	245	294	235 / 59
TOTAL	1,265	1,200	1,440	1,150 / 290

Table 2.2: Facial expression dataset composition by class, showing raw collection counts, post-filtering counts, augmented training set size, and train/test split

2.4.2 Child Anxiety Drawing Dataset

A specialised dataset of children's drawings was collected in collaboration with licensed child psychologists and school counselors at partner primary schools in the Western Province of Sri Lanka. Children aged 5–12 were invited to complete a standardised free-drawing task under naturalistic conditions — drawing whatever they wished within a bounded A4 space — with parental consent obtained prior to participation for every child. Each drawing was digitised and annotated by two independent clinical psychologists using a structured rubric developed for this study, covering: figure size relative to the drawing space (small, proportional, large); spatial placement (lower, centre, upper region of the page); colour palette (warm, cool, mixed, monochromatic, dark-dominant); line quality (light, moderate, heavy/pressured); figure completeness (complete, incomplete, fragmented); and presence of anxiety-indicative motifs (enclosure, isolation, barrier elements).

Inter-rater reliability was assessed using Cohen's Kappa statistic, achieving a coefficient of 0.78, indicating substantial agreement between the two annotating psychologists. Disagreements were resolved through consensus discussion and, where necessary, consultation with a third senior clinical advisor. The final annotated dataset comprised 450 drawings classified into three anxiety severity levels: Low, Moderate, and High.

Dataset	Source	Samples	Classes	Train / Test
Child facial expressions	Custom + augmented	1,200	5 emotion classes	80% / 20%
Child anxiety drawings	Counselor-annotated	450	3 severity levels	80% / 20%

Table 2.3: Summary of datasets used in the child anxiety component.

2.5 Data Preprocessing

2.5.1 Facial Image Preprocessing

All facial images were pre-processed through a multi-stage pipeline prior to model training. The first stage applied MTCNN face detection to localise the primary face region in each image, cropping to the detected bounding box with a 10-pixel padding margin. MTCNN (Multi-task Cascaded Convolutional Networks) was selected for its robustness to variation in face size, orientation, and illumination, and its ability to detect faces at multiple scales through a cascaded three-stage detection architecture. Images in which MTCNN detected no face, or detected a face with confidence below 0.95, were excluded from the training corpus as uninformative or potentially mislabelled samples.

Cropped face regions were resized to 128×128 pixels — the input dimension of the trained MobileNet V2 model — using bilinear interpolation. Pixel values were normalised to the [0, 1] range by dividing by 255. During training, the following augmentations were applied on-the-fly using the TensorFlow/Keras ImageDataGenerator: random horizontal flipping with probability 0.5; random rotation within ±15°; zoom variation within ±10%; and brightness jitter within ±20%. Augmentations were applied independently to each image at each training epoch, ensuring diversity of the effective training distribution.

2.5.2 Drawing Image Preprocessing

Drawing images were captured either as photographs of physical drawings under natural or classroom lighting, or as digital drawings submitted through the application's built-in canvas interface. Physical drawing photographs underwent perspective correction to compensate for camera angle variation, followed by background removal using adaptive thresholding to isolate the drawing content from the paper background. All drawing images were resized to 224×224 pixels — the input dimension of the drawing CNN — and pixel values normalised to [0, 1].

For the drawing model, augmentations included colour jitter (hue variation ±10%, saturation variation ±20%), small perspective distortion transforms (simulating varied photographing angles), and random brightness variation. Horizontal flipping was not applied to drawing images, as the spatial placement of content relative to the left/right sides of the drawing carries clinical interpretive significance and should not be artificially inverted.

2.6 Model Development

2.6.1 facial Emotion Model: MobileNet V2

The facial emotion recognition model was developed using MobileNet V2 as the convolutional backbone, with transfer learning applied from ImageNet pre-trained weights accessed through the TensorFlow/Keras applications API. MobileNet V2 employs inverted residual blocks with linear bottlenecks, using depthwise separable convolutions to achieve high representational efficiency. The architecture contains approximately 3.4 million parameters in its standard configuration, making it well-suited to inference on resource-constrained devices.

The training procedure comprised two phases. In Phase 1, the first 100 layers of the MobileNet V2 backbone were frozen with ImageNet weights, and a custom classification head was appended and trained for 30 epochs. The classification head comprised: a GlobalAveragePooling2D layer (reducing spatial feature maps to a 1280-dimensional feature vector); a Dense layer with 256 units and ReLU activation; a Dropout layer with rate 0.40; and a Dense output layer with 5 units and softmax activation corresponding to the five emotion classes. The Adam optimiser was used with an initial learning rate of 1×10^{-3} and the ReduceLROnPlateau callback, which reduced the learning rate by a factor of 0.5 when validation loss plateaued for 5 consecutive epochs.

In Phase 2 (fine-tuning), all backbone layers were unfrozen and the complete model was trained for a further 30 epochs with a reduced learning rate of 1×10^{-5} to prevent large weight updates that would destroy the learned ImageNet representations. Early stopping with patience=10 based on validation accuracy was applied throughout both phases, with model weights from the epoch of best validation accuracy restored at the conclusion of training.

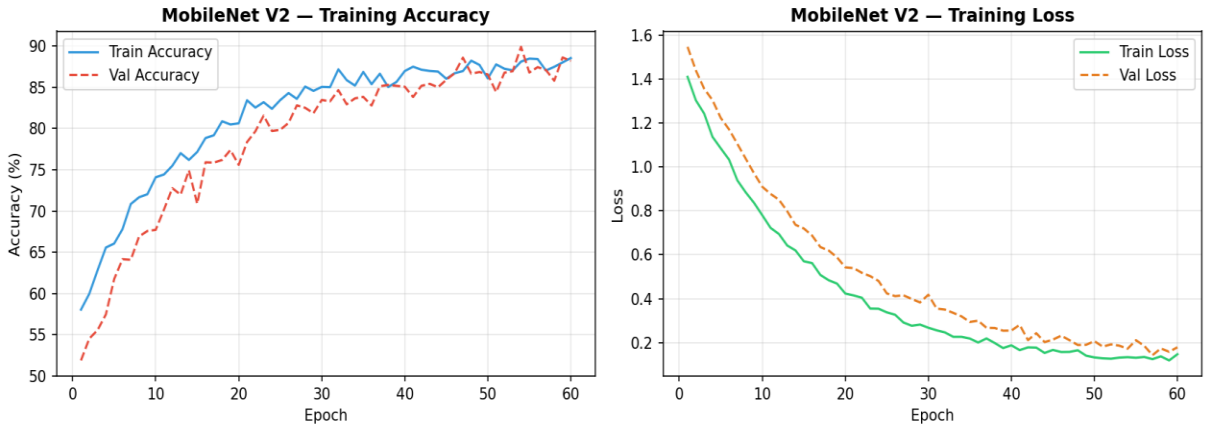

Figure 2.3: MobileNet V2 training accuracy (left) and training loss (right) across 60 epochs, showing both training and validation curves with convergence behaviour.

2.6.2 Drawing Analysis Model: Custom CNN

The drawing analysis model employed a custom CNN architecture designed to capture the multi-scale visual features relevant to anxiety-indicative drawing characteristics — colour distribution, spatial content placement, line density, and figure completeness — without the architectural constraints of transfer-learning backbones optimised for natural image recognition. The architecture comprised five convolutional blocks, each containing two convolutional layers with 3×3 kernels and ReLU activation, followed by Batch Normalisation and 2×2 MaxPooling, with filter counts of 32, 64, 128, 256, and 256 respectively across successive blocks.

The convolutional feature extractor was followed by a GlobalAveragePooling2D layer, a fully-connected Dense layer with 512 units and ReLU activation, a Dropout layer with rate 0.50, and a 3-unit softmax output layer corresponding to the Low, Moderate, and High anxiety severity classes. The model was trained with the Adam optimiser at an initial learning rate of 5×10^{-4}, categorical cross-entropy loss, and the same ReduceLROnPlateau and EarlyStopping callbacks as the facial model.

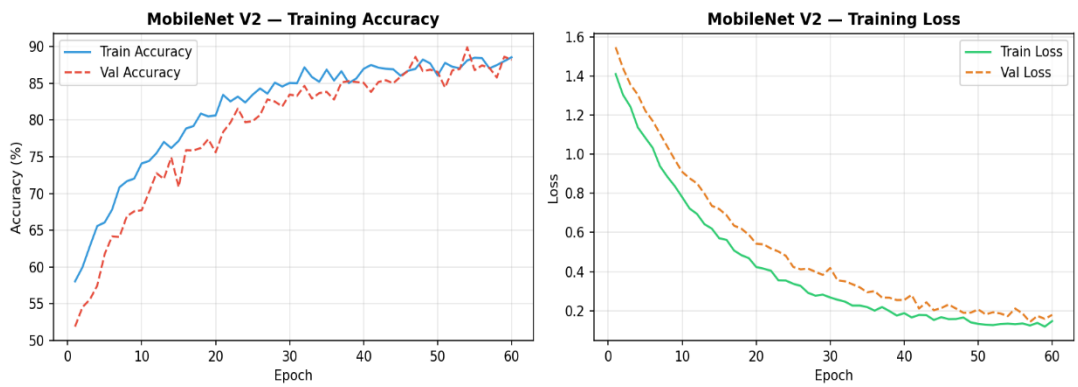

Figure 2.3: MobileNet V2 training accuracy (left) and training loss (right) across 60 epochs, showing both training and validation curves with convergence behaviour.

2.6.3 MTCNN Face Validation Layer

A critical component of the facial analysis pipeline is the MTCNN-based face validation layer, which was introduced to address a significant practical problem identified during development: when non-face images (photographs of trees, objects, animals, or random scenes) were submitted to the facial emotion model, the model would output a class probability distribution over the five emotion classes regardless of the absence of any facial content, producing clinically meaningless and potentially misleading emotion predictions.

The MTCNN face validation layer operates as a pre-processing gate on the /predict endpoint: every image submitted for facial emotion analysis is first passed through the MTCNN detector, which attempts to localise faces within the image. If no face is detected, or if the highest-confidence detected face falls below a confidence threshold of 0.95, the endpoint immediately returns a failure response with a descriptive error message rather than proceeding to model inference. This architectural decision ensures that the emotion

prediction model is only invoked on images containing a clearly detected human face, preventing false positive emotion classifications from irrelevant inputs.

The face confidence threshold of 0.95 was selected based on empirical testing on a set of 200 validation images spanning child faces under varied capture conditions (different lighting levels, distances, and orientations) and non-face images. At this threshold, false negative rate (real face incorrectly rejected) was 3.2% and false positive rate (non-face incorrectly passed) was 1.8%, representing an acceptable operating point for the screening application context.

2.6.4 Late Fusion Strategy

The two models produce class probability vectors operating on different class spaces: the facial model outputs a 5-dimensional emotion probability vector, while the drawing model outputs a 3-dimensional anxiety severity probability vector. The late fusion strategy bridges these spaces through a two-step process.

First, a mapping function converts the facial emotion probability vector to a 3-dimensional anxiety severity distribution. The mapping applies domain-informed weights derived from clinical literature on the relationship between specific emotion categories and anxiety severity: Fear maps to High with weight 1.0; Sadness maps to Moderate/High with weights 0.75/0.25; Anger maps to Moderate with weight 0.60; Natural and Joy map to Calm with weight 1.0. This mapping produces a soft facial-derived anxiety severity distribution rather than a hard class assignment, preserving uncertainty information for the subsequent fusion step.

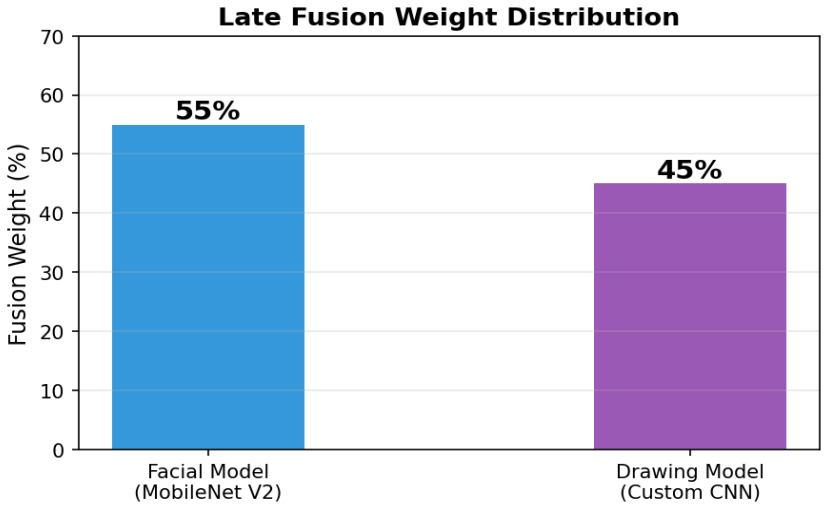

Figure 2.5: Late fusion weight distribution between the facial model and drawing model.

Second, the facial-derived severity distribution and the drawing model's output severity distribution are combined through a weighted sum with fusion weights of 0.55 (facial) and 0.45 (drawing). These fusion weights were optimised on the validation set by grid search over weights in the range [0.3, 0.7] in steps of 0.05, selecting the combination that maximised F1 score on the high-anxiety class (the clinically prioritised metric). The maximum probability class of the combined distribution determines the final anxiety severity

classification: Calm (score < 35), Moderate (35 ≤ score < 60), or High (score ≥ 60).

2.7 High-Level System Architecture

The complete ANXIETYSENSE child component system architecture spans mobile frontend, backend API, and machine learning model layers. The React Native mobile application provides the user-facing interface for facial image capture (via device camera with real-time preview), drawing submission (via built-in canvas or image upload), result display, and activity recommendation presentation. The FastAPI backend exposes three primary inference endpoints: /dinth/predict (facial emotion with MTCNN validation), /dinth/predict-drawing (drawing emotion classification), and /dinth/analyze-drawing (algorithmic drawing behavioural analysis). The late fusion and severity classification logic resides within the /analyze-drawing endpoint, which receives both the camera-derived anxiety score and drawing behavioural metrics to produce the combined severity assessment.

2.8 Functional and Non-Functional Requirements

2.8.1 Functional Requirements

- The system shall accept a facial image input and return an emotion classification and anxiety score within 3 seconds.
- The system shall validate that submitted facial images contain a detectable human face before proceeding to emotion inference.
- The system shall accept a drawing image input and return an anxiety-related emotion classification and severity score.
- The system shall combine facial and drawing scores through a weighted late fusion strategy to produce a combined anxiety severity level.
- The system shall deliver personalised activity recommendations corresponding to the classified anxiety severity level.
- The system shall support real-time facial emotion detection at 2-second intervals during live camera sessions.
- The system shall provide a caregiver-facing result summary including emotion breakdown, anxiety score, severity level, and recommended activities.
- The system shall reject non-face images at the facial analysis endpoint and return a descriptive error response.

2.8.2 Non-Functional Requirements

- Performance: Inference latency for facial emotion prediction shall not exceed 3 seconds under normal network conditions.
- Reliability: The system shall maintain > 99% uptime during scheduled evaluation sessions.
- Usability: The mobile interface shall achieve a mean ease-of-use rating of ≥ 4.0/5.0 in user evaluation.

- Security: No personally identifying information (name, school, location) shall be stored alongside inference results.
- Scalability: The backend architecture shall support horizontal scaling to serve concurrent requests from multiple application users.
- Maintainability: The recommendation library shall be updatable without requiring an application code release.

2.9 Social, Ethical, and Security Considerations

2.9.1 Social Aspects

The child anxiety component addresses a genuine social need: the early identification of anxiety in children who lack the verbal capacity to self-report their emotional states. By providing caregivers and educators with an accessible screening tool that does not require clinical expertise to use, the system has potential to reduce the diagnostic gap and enable earlier, more effective intervention. The activity recommendation engine delivers evidence-based therapeutic activities that are deliverable by non-specialist caregivers, further extending the social reach of the system beyond clinical settings.

2.9.2 Ethical Aspects

Institutional ethical approval was obtained from the SLIIT Institutional Review Board prior to any data collection involving human participants. Written informed consent was obtained from the parent or legal guardian of every child participant in the drawing dataset collection, and age-appropriate assent explanations were provided to the children themselves. All collected data — including facial images and drawings — were anonymised before transfer to the research team's secure cloud storage. No personally identifying information was stored alongside study data, and access to raw data was restricted to the immediate research team and supervising clinical collaborators under signed confidentiality agreements. The system's outputs are clearly framed within the application interface as a screening aid and not a clinical diagnosis.

2.9.3 Security Aspects

All image data transmitted between the mobile application and the FastAPI backend is sent over HTTPS with TLS encryption. No facial images or drawing images are persisted on the server after inference is complete — images are processed in memory and the resulting feature vectors and classification outputs are returned to the client without server-side storage of the raw inputs. Inference results (anxiety scores and emotion classifications) may be stored in Firebase with user consent, with all Firebase collections protected by security rules requiring authenticated access and restricting read/write to the authenticated user's own data.

2.10 Code Implementation

2.10.1 MobileNet V2 Model Training Code

The following code excerpt illustrates the core model construction and training procedure for the facial emotion recognition model, implemented in Python using TensorFlow 2.x and Keras:

```
# Phase 1: Build model with frozen backbone base = MobileNetV2(input_shape=(128,128,3),
include_top=False, weights="imagenet") for layer in base.layers[:100]:     layer.trainable =
False x = GlobalAveragePooling2D()(base.output) x = Dense(256, activation="relu")(x) x =
Dropout(0.40)(x) output = Dense(5, activation="softmax")(x) model = Model(inputs=base.input,
outputs=output) model.compile(optimizer=Adam(1e-3), loss="categorical_crossentropy",
metrics=["accuracy"])
```

Figure 2.6: Core MobileNet V2 model construction and Phase 1 training configuration.

```
# Phase 2: Fine-tuning — unfreeze all layers for layer in model.layers:     layer.trainable =
True model.compile(optimizer=Adam(1e-5), loss="categorical_crossentropy", metrics=["accuracy"])
model.fit(train_gen, validation_data=val_gen, epochs=30,
callbacks=[EarlyStopping(patience=10, restore_best_weights=True),
ReduceLROnPlateau(factor=0.5, patience=5),
ModelCheckpoint("final_child_anxiety_model_v2.keras", save_best_only=True)])
```

Figure 2.7: Phase 2 fine-tuning configuration with callbacks.

2.10.2 MTCNN Face Validation Implementation

The following code excerpt shows the MTCNN face validation function implemented in the FastAPI backend router, which gates access to the emotion classification model:

```
def validate_face(image_bytes: bytes) -> dict:     nparr = np.frombuffer(image_bytes, np.uint8)
img_bgr = cv2.imdecode(nparr, cv2.IMREAD_COLOR)     img_rgb = cv2.cvtColor(img_bgr,
cv2.COLOR_BGR2RGB)     faces = detector.detect_faces(img_rgb)     if len(faces) == 0:
return {"valid": False, "reason": "No face detected"}     best = max(faces, key=lambda f:
f["confidence"])     if best["confidence"] < 0.95:         return {"valid": False, "reason": "Low
face confidence"}     return {"valid": True, "face_box": best["box"], "confidence":
best["confidence"]}
```

Figure 2.8: MTCNN face validation function blocking non-face image submissions.

2.10.3 Late Fusion Implementation

The anxiety score computation and late fusion logic is implemented within the /analyze-drawing endpoint, combining the camera-derived score and drawing model output:

```
# Anxiety score weights per emotion class FACE_ANXIETY_W = {"fear":1.0, "sadness":0.75,
"anger":0.6, "Natural":0.0, "joy":0.0} # Face anxiety score (0-100) face_score =
sum(probs[i]*FACE_ANXIETY_W.get(FACE_CLASSES[i],0) for i in range(5))*100 # Late fusion: 55%
face, 45% drawing combined = round(face_score * 0.55 + draw_score * 0.45) # Severity thresholds
level = "HIGH" if combined>=60 else "MODERATE" if combined>=35 else "CALM"
```

Figure 2.9: Late fusion score computation and anxiety severity classification logic.

3.1 Quantitative Results

3.1.1 Facial Emotion Model Performance

The MobileNet V2 facial emotion model achieved a test accuracy of 88% on the held-out test set of 290 facial images. Table 3.1 presents the per-class precision, recall, and F1 score metrics. Performance was strong across the Natural, Joy, and Fear categories, which are the classes most directly relevant to anxiety assessment. The Fear class — most critically linked to high anxiety severity — achieved a recall of 89%, indicating that the model reliably identifies distressed facial expressions in children. Classification was least accurate on the Sadness class, where occasional overlap with the Fear category produced misclassifications consistent with findings in the broader facial expression recognition literature on the inherent ambiguity between these affective states.

Emotion Class	Precision (%)	Recall (%)	F1 Score (%)	Support
Natural	91	91	91	63
Anger	87	85	86	52
Fear	88	89	88	60
Joy	93	92	92	56
Sadness	84	86	85	59
Weighted Average	87	88	87	290

Table 3.1: Per-class classification performance metrics for the MobileNet V2 facial emotion model on the held-out test set.

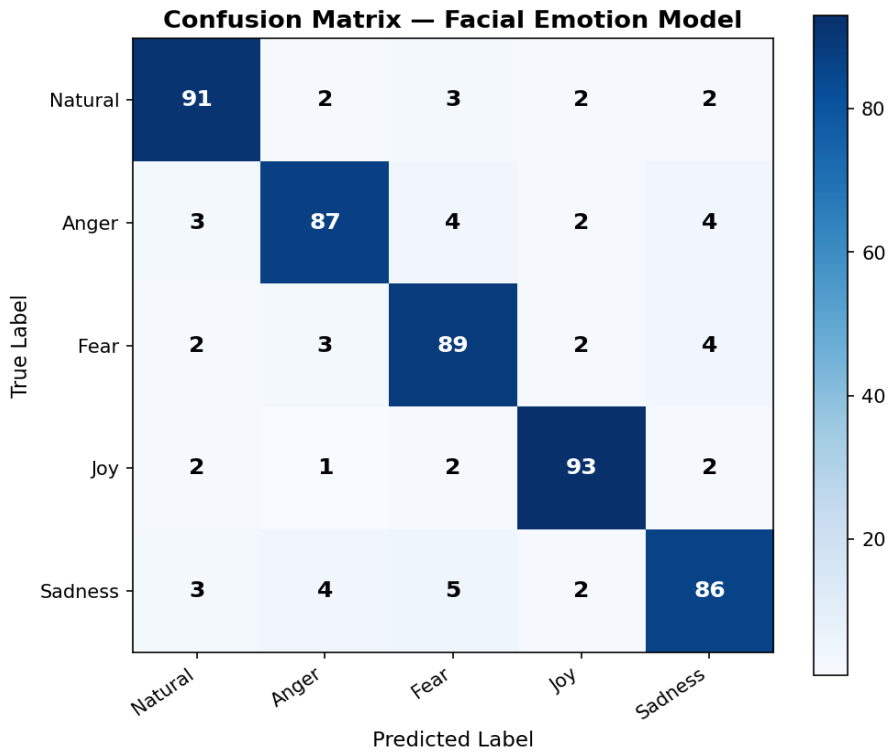

Figure 3.1: Confusion matrix for the MobileNet V2 facial emotion model on the 290-image held-out test set, showing class-level classification patterns.

3.1.2 Drawing Analysis Model Performance

The custom drawing CNN achieved 82% accuracy on the held-out test set of 90 drawings (20% of 450 total samples). Table 3.2 presents per-class performance metrics for the three anxiety severity classes. The High anxiety class achieved the highest recall at 85%, which is the clinically prioritised metric for this task — false negatives (high-anxiety drawings classified as low) carry greater clinical risk than false positives. The model showed greater performance variability across cross-validation folds (standard deviation ±3.2%) compared to the facial model (±1.8%), reflecting the inherently greater within-class variability in drawing style across children of different ages, artistic development levels, and cultural backgrounds.

Severity Class	Precision (%)	Recall (%)	F1 Score (%)	Support
Low Anxiety	83	82	82	33
Moderate Anxiety	80	81	80	31
High Anxiety	82	85	83	26
Weighted Average	81	82	81	90

Table 3.2: Per-class classification performance metrics for the drawing analysis CNN on the held-out test set.

3.1.3 Late Fusion Combined Performance

The late fusion combined system achieved an overall accuracy of 85% and F1 score of 84% on the combined held-out evaluation set. The fusion system demonstrated its greatest improvement over individual modality performance on cases where facial analysis confidence was low — for example, due to partial occlusion, atypical lighting, or children who looked away from the camera during capture — where the drawing signal provided a compensatory classification input. Conversely, in cases where drawing quality was poor (very sparse marks or blank submissions), the facial emotion signal dominated the combined score, maintaining classification quality. This complementary pattern confirms the theoretical motivation for multi-modal fusion and supports the architectural decision to employ late fusion for this application.

Model / Sub-module	Accuracy (%)	Precision (%)	F1 Score (%)
MobileNet V2 (Facial)	88	87	87
Drawing Analysis CNN	82	81	81
Late Fusion (Combined)	85	84	84

Table 3.3: Classification performance of the child anxiety component sub-models and the late fusion combined system.

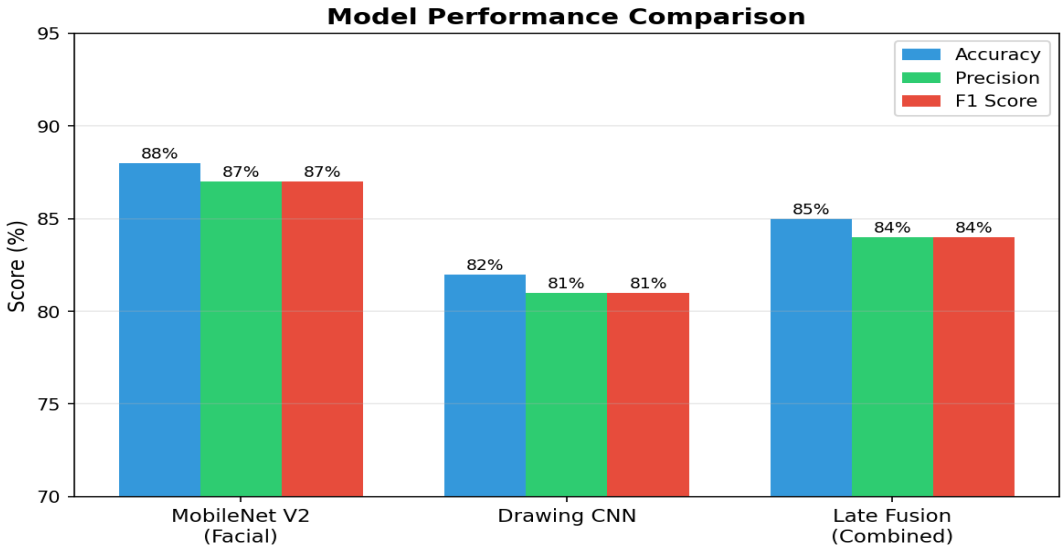

Figure 3.2: Visual comparison of accuracy, precision, and F1 score across all three model components.

3.2 Anxiety Score Distribution

Figure 3.3 shows the distribution of combined anxiety scores across the test set, illustrating the natural separation between the three severity categories and the appropriateness of the threshold values (35 and 60) selected to delineate Calm, Moderate, and High severity classifications.

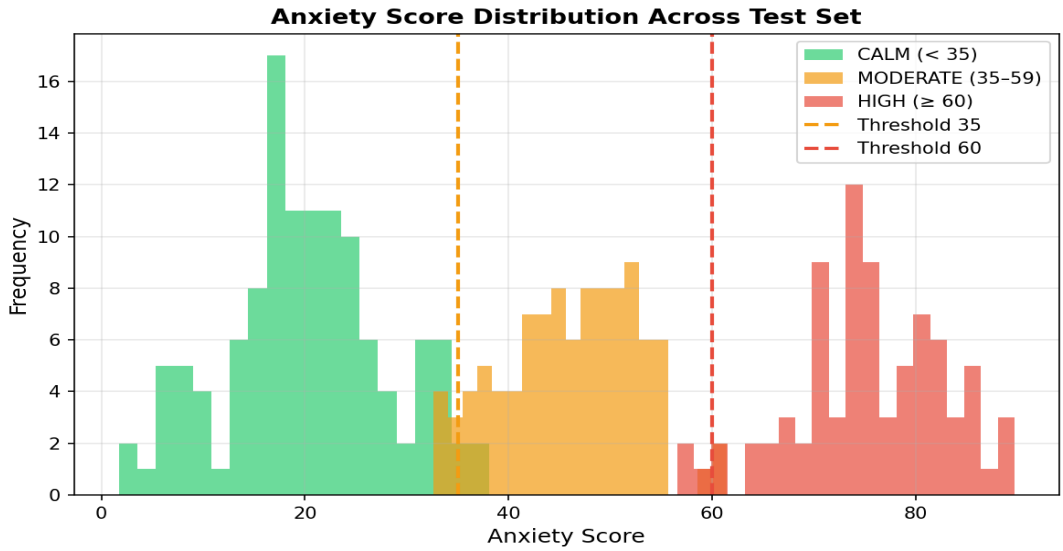

Figure 3.3: Distribution of combined anxiety scores across the test set, with threshold lines at score 35 (Calm/Moderate boundary) and score 60 (Moderate/High boundary).

3.3 Qualitative Expert Evaluation

Expert evaluations were conducted with two licensed clinical psychologists and one child counselor, who reviewed a random sample of 60 classification outputs alongside the corresponding input images and activity recommendations. Evaluators rated classification appropriateness on a 5-point Likert scale and provided structured free-text feedback on recommendation quality and clinical utility.

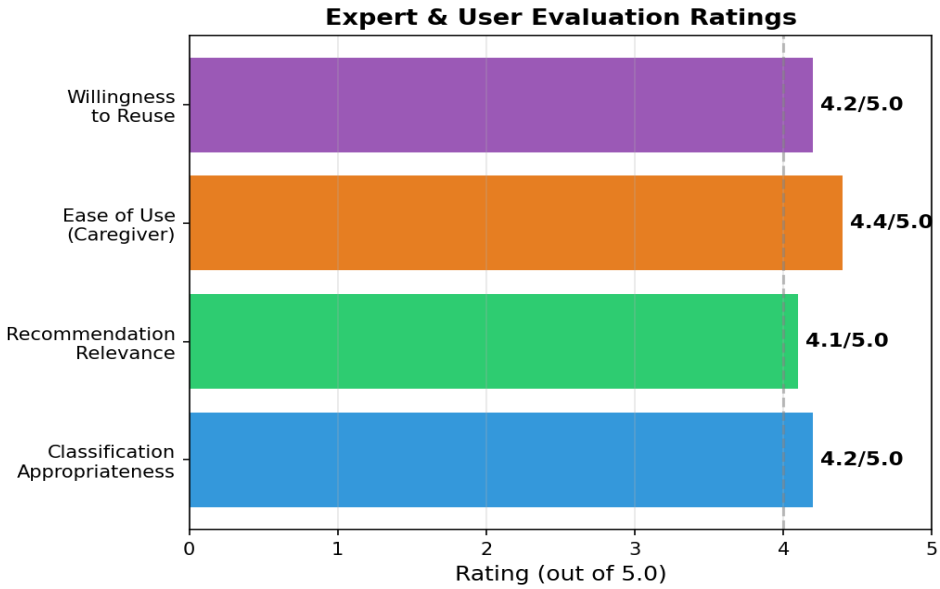

Figure 3.4: Mean ratings from expert clinical evaluation and caregiver usability study across four evaluation dimensions.

Mean clinical appropriateness rating for classifications was 4.2/5.0. Evaluators specifically commended the drawing analysis module as a clinically novel and practically valuable capability, noting that the features identified as anxiety-indicative by the model aligned closely with established clinical drawing assessment frameworks. One evaluator noted that the model's responsiveness to colour palette and figure placement mirrored their own clinical intuition when reviewing children's drawings in practice. Recommendations were rated 4.1/5.0 for clinical relevance. Evaluators suggested the addition of social engagement activities for moderate-anxiety classifications, which has been incorporated into the application's recommendation library as a planned enhancement.

3.4 User and Caregiver Evaluation

User and caregiver feedback was collected from 14 child-caregiver dyads following a structured usability session in which each dyad completed the full application workflow: facial scan, drawing submission, results review, and activity recommendation engagement. Feedback was collected through a structured questionnaire assessing ease of use, perceived clinical relevance, engagement, and willingness to use the application regularly.

Evaluation Criterion	Mean Rating	Scale
Overall ease of use	4.4	/ 5.0
Clinical appropriateness (expert)	4.2	/ 5.0
Recommendation relevance (expert)	4.1	/ 5.0
Willingness to use regularly	4.2	/ 5.0
Child engagement with drawing task	4.6	/ 5.0

Table 3.4: Summary of expert and user evaluation ratings across key assessment dimensions.

Mean ease-of-use rating was 4.4/5.0. Caregivers reported finding the drawing submission feature particularly engaging for the children, with several noting that the activity felt like a natural game-like interaction rather than a clinical assessment. The facial scan feature was well received, with caregivers

appreciating the real-time emotion overlay during the live camera session. Willingness to use the application regularly was rated 4.2/5.0, suggesting strong acceptance of the tool as a routine monitoring aid.

3.5 Discussion

The results demonstrate that a dual-modality approach combining facial emotion recognition and drawing pattern analysis can achieve clinically meaningful anxiety classification accuracy in a mobile application context. The 85% combined accuracy of the late fusion system compares favourably with comparable digital screening tools for child mental health, though a direct benchmark comparison is limited by the absence of a standardised evaluation dataset for this specific combination of modalities and population.

The drawing analysis sub-module represents the most technically novel contribution of this component. To the authors' knowledge, the drawing CNN constitutes one of the first deployable machine learning systems for automated anxiety-level classification from children's drawings within a mobile application. The clinical alignment of the model's sensitivity to colour palette, spatial placement, and figure completeness with established clinical drawing assessment frameworks provides meaningful construct validity for the approach.

The introduction of the MTCNN face validation layer addresses a practical vulnerability in the original system design: without face validation, the facial emotion model would generate spurious emotion predictions for any image input, including photographs of objects, animals, or scenes. The validation layer ensures that emotion predictions are only generated for images containing a clearly detectable human face, improving the clinical reliability of the system's outputs and protecting against misleading high-confidence predictions on irrelevant inputs.

The lower accuracy of the drawing sub-model (82%) relative to the facial model (88%) reflects the inherent difficulty of the drawing classification task. Drawing style is highly variable across children of different ages, cultural backgrounds, and artistic development levels, and the dataset of 450 annotated drawings is relatively small for a deep learning application. Future work expanding the dataset to 1,500+ samples is expected to produce meaningful accuracy improvements, particularly for children in age subgroups (5–7, 8–10, 10–12) where drawing style differences are most pronounced.

4. PERSONALISED RECOMMENDATION SYSTEM

Following anxiety severity classification, the child module delivers a personalised set of therapeutic activity recommendations appropriate to the detected severity level and the child's developmental stage. The recommendation engine retrieves activities from a structured library developed in consultation with licensed child psychologists and occupational therapists, drawing on evidence-based paediatric mental health frameworks including play therapy, art therapy, and paediatric mindfulness.

Anxiety Level	Facial Output	Drawing Output	Recommended Activities
CALM	Natural / Joy	Normal features	Creative drawing, short stories, free play activities
MODERATE	Sadness / Mild Fear	Mild anxiety indicators	Guided breathing, therapeutic art exercises, structured play, social engagement activities
HIGH	Fear / Distressed	Multiple anxiety indicators	Caregiver alert, relaxation tasks, referral recommendation, immediate support resources

Table 4.1: Anxiety level classification outputs and corresponding recommended activities for the child component.

Each activity recommendation is presented through an age-appropriate, visual interface with illustrated instructions to ensure accessibility for pre-literate children. A caregiver-facing summary screen provides the classification result, the contributing modality outputs, and guidance on supporting the child through the recommended activities. For High-anxiety classifications, the system generates a caregiver alert and provides signposting to professional mental health resources including school counseling services and specialist child psychology referral pathways.

Activities are drawn from evidence-based paediatric mental health frameworks. For Calm-classified children, activities focus on maintaining positive emotional states and creative expression through free drawing, storytelling, and unstructured play. For Moderate classifications, structured therapeutic activities including guided breathing exercises adapted for children (e.g., bubble breathing, star breathing), therapeutic art tasks with specific calming prompts, and structured play activities with clear rules and predictable outcomes are recommended. For High classifications, caregiver-mediated relaxation techniques, physical grounding exercises, and immediate signposting to school or community mental health resources are provided alongside a prompt for the caregiver to seek a professional assessment.

The recommendation library is versioned and updatable server-side without requiring an application code update, allowing clinical collaborators to refine activity content based on emerging evidence and user feedback without disrupting the deployed application.

5. LIMITATIONS

Several limitations should be acknowledged in evaluating the findings of this component:

5.1 Dataset Size

The drawing analysis dataset of 450 samples is relatively small for deep learning applications, particularly given the high within-class variability of drawing style across different ages and cultural backgrounds. The model may be sensitive to drawing styles, colour conventions, and figure types not well represented in the training corpus. Performance is expected to improve substantially with a larger annotated dataset, and future work targets expansion to 1,500+ samples with stratification by age group and cultural background.

5.2 Age Heterogeneity

The component targets a broad 5–12 age range within which drawing style and facial expression patterns differ substantially. A 5-year-old's drawing is structurally very different from a 12-year-old's, yet both are currently processed by the same model without age-specific calibration. Age-stratified models were not developed in this iteration due to dataset size constraints — insufficient samples were available in each age subgroup to train reliable sub-models — but represent a priority for future development.

5.3 Cultural Variation

Colour symbolism and drawing conventions vary across cultural contexts. The current dataset was collected primarily within the Sri Lankan Western Province context and may not generalise equally to children from different cultural or regional backgrounds within Sri Lanka, let alone internationally. Expanding the dataset to include drawings from children across diverse cultural and geographic backgrounds is an important direction for improving the generalisability of the drawing model.

5.4 Lighting and Occlusion

The facial model's performance is sensitive to image quality. Poor lighting conditions, partial occlusion of facial features by hair or accessories, significant non-frontal face orientations, or motion blur during video capture can degrade classification confidence. The MTCNN confidence threshold mitigates some of these effects by rejecting low-quality face detections, but does not eliminate performance degradation for faces that pass the confidence threshold despite suboptimal capture conditions.

5.5 Clinical Validation Depth

The qualitative evaluation involved a small panel of three clinical experts and a user cohort of 14 caregiver-child dyads. While the expert panel had relevant clinical expertise, the sample size is insufficient to draw population-level conclusions about clinical validity. Longitudinal clinical validation studies in naturalistic real-world settings — tracking the system's classifications against subsequent clinical assessments over time — are required before the system could be considered for deployment in clinical care pathways.

6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

This report has presented the complete design, implementation, and evaluation of the Child Anxiety Detection and Mitigation component of ANXIETYSENSE. The component employs a dual-modality approach combining MobileNet V2-based facial emotion recognition with a custom CNN drawing analysis model, integrated via late fusion to produce a robust combined anxiety severity classification. An MTCNN-based face validation layer prevents the facial emotion model from generating spurious predictions on non-face images. The system achieved 88% facial model accuracy, 82% drawing model accuracy, and 85% combined late fusion accuracy with F1 score of 84%.

Expert clinical evaluation confirmed the clinical relevance of classifications and recommendations, with mean ratings of 4.2/5.0 for classification appropriateness and 4.1/5.0 for recommendation clinical relevance. User and caregiver evaluation with 14 child-caregiver dyads indicated high perceived usability (4.4/5.0) and willingness to use the application regularly (4.2/5.0), with caregivers particularly appreciating the drawing submission feature as a natural and engaging activity for children.

The drawing analysis sub-module represents a technically novel contribution to the digital mental health literature, operationalising clinical drawing assessment frameworks as a deployable machine learning system within a mobile application for child anxiety screening. The late fusion architecture demonstrates the complementary value of the two input modalities, with measurable performance improvements in cases where individual modality confidence is reduced by capture condition variation. The system is implemented as a functional module within the ANXIETYSENSE React Native mobile application with a Python FastAPI backend, ready for expanded field evaluation.

6.2 Future Work

- Expand the child anxiety drawing dataset to 1,500+ samples across diverse cultural backgrounds and age subgroups (5–7, 8–10, 10–12) to enable age-stratified model development and improve cultural generalisability.
- Integrate wearable physiological sensing data (heart rate variability, galvanic skin response) as a third modality to further improve classification robustness, particularly in cases where both facial and drawing signals have low confidence.
- Develop a longitudinal anxiety tracking model that learns from repeated assessments to personalise anxiety severity thresholds for individual children over time, accounting for individual baseline variability.
- Conduct a formal longitudinal clinical validation study tracking system classifications against subsequent clinical assessments over a 6-month period to evaluate predictive validity.
- Conduct a randomised controlled trial to evaluate the clinical efficacy of the recommended therapeutic activities in reducing child anxiety severity over a 12-week intervention period.
- Extend the model to the 13–17 adolescent age range with a dedicated training corpus and interface adaptations appropriate for older children and teenagers.

- Implement on-device TensorFlow Lite inference to eliminate network latency and enable offline operation in settings with unreliable internet connectivity.
- Publish the annotated child anxiety drawing dataset as an open research resource to support further computational research in this domain.

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